Langqi Xing

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EDUCATION

Northeastern University Boston, MA
Doctor of Philosophy in Mechanical Engineering Expected Graduation: Dec 2027
Relevant Courses: Multiphase Transport, Transport Phenomena, Mathematical Methods

University of California, Santa Barbara Santa Barbara, CA
Bachelor of Science in Chemical Engineering Dec 2022
Relevant Courses: Separation Processes, Linear Algebra, Multi-variable Calculus, Soft Materials

SKILLS

Data Analysis: MATLAB, Python, ImageJ, Aspen HYSYS, SolidWorks, COMSOL, Power BI
Experimental Design: Microfluidic, 3-D printing, Confocal Microscope, Microparticle Fabrication
Languages: English (Fluent), Mandarin (Native)

RESEARCH EXPERIENCE

Graduate Research Assistant, Northeastern University Jan 2023 – Present
Prof. Xiaoyu Tang's Multiphase Lab

- Designed and fabricated microfluidic dead-end channels; patterned NOA-81 UV adhesive to create symmetric/asymmetric boundary materials (SolidWorks, Soft Lithography)
- Conducted micro-particle transport experiments under controlled solute/surfactant gradients to achieve pump-free particle delivery
- Quantified particle trajectories, velocity components, and positions analytically (MATLAB, ImageJ)
- Developed and validated analytical/simulation models to predict concentration fields and boundary-driven backflow across materials (COMSOL)

Undergraduate Research Assistant, UCSB Dec 2020 – Dec 2022
Prof. Alban Sauret's Fluid Lab

- Measured velocity profiles of capillary flows of complex suspensions
- Analyzed fiber counting/orientation and granular flow (MATLAB, ImageJ)
- Investigated rheological behavior of polystyrene suspensions confined between parallel plates
- Performed entrainment experiments of fibers with varying diameters/lengths on rod substrates

Undergraduate Research Assistant, UCSB Nov 2021 – Dec 2022
C-Thru Global Project with Profs. Phillip Christopher and Eric Masanet

- Quantified energy/emissions footprints of global chemical supply chains
- Conducted literature reviews and constructed mass/energy balance models of chemical processes

Undergraduate Research Assistant, UCSB Dec 2020 – Jun 2021
Prof. Eric McFarland's Energy Lab

- Performed heat conduction/convection calculations for filament reactors
- Created COMSOL Multiphysics models and analyzed reactor temperature profiles

PUBLICATIONS

- **Controlling particle dynamics in dead-end channels via boundary effects** Apr 2026
  L. Xing, X. Tang.
  Nanoscale, Published
- **Effects of interparticle cohesion on the collapse of granular columns** Jul 2024
  D.-H. Jeong, **L. Xing**, M. Ka Ho Lee, N. Vani, A. Sauret.
  Physical Review Fluids, Published

- **Deposition and alignment of fiber suspensions by dip coating** Jun 2023
  D.-H. Jeong, **L. Xing**, M. Ka Ho Lee, N. Vani, A. Sauret.
  Journal of Colloid and Interface Science, Published
- **Particulate suspension coating of capillary tubes** Oct 2022
  D.-H. Jeong, **L. Xing**, J.-B. Boutin, A. Sauret.
  Soft Matter, Published

CONFERENCES

Engineering tunable hydrogel beacon configurations for extended-range colloidal transport
2025 AIChE Annual Meeting, Boston, MA Nov 2025
- Presented a microfluidic strategy to generate sustained electrolyte gradients for long-range particle delivery into dead-end channels
- Quantified boundary-material-dependent diffusio-osmotic backflow and separatrix shifts to map transport regimes and inform device design

Controlling particle dynamics in dead-end channels via tunable boundary effects
77th Annual Meeting of the Division of Fluid Dynamics, Salt Lake City, UT Nov 2024
- Explored diffusiophoresis-driven colloidal transport under electrolyte gradients, focusing on boundary-driven fluid flows
- Demonstrated manipulation of particle trajectories by modifying wall streaming potentials at dead-end boundaries

Leveraging boundary effects for particle delivery into a dead-end channel
99th New England Complex Fluids Workshop, Brown University, Providence, RI Jun 2024
- Presented microfluidic approaches to tune streaming potentials and achieve controlled colloidal motion

POSTERS

Controlling particle dynamics in dead-end channels via tunable boundary effects
AAAS Annual Meeting, Boston, MA Feb 2025
- Delivered an interactive e-poster to a broad scientific audience and judges on boundary-tuned particle delivery in dead-end microchannels
- Discussed how different boundary materials shift particle turning behaviors, supported by experiments and theory

Engineering tunable hydrogel beacon configurations for extended-range colloidal transport
Northeastern University MIE Graduate Research & Art Expo, Boston, MA Dec 2025
- Presented a research poster (and associated visualization materials) on “chemical beacon” designs that generate sustained gradients for pump-free particle delivery
- Communicated key mechanisms and performance trends using publication-quality graphics and flow visualizations

Controlling particle dynamics in dead-end channels via tunable boundary effects
Northeastern University College of Engineering Research Expo, Boston, MA Oct 2024
- Showcased a microfluidic platform and modeling framework for boundary-driven particle transport; engaged faculty and student researchers on applications and next steps

WORK EXPERIENCE

R&D Trainee, PepsiCo Jul 2021 – Sep 2021
- Understood ingredient-type landscape and collected existing ingredient specifications
- Leveraged AI techniques (Excel, Python Modules, Power BI) to standardize parameters
- Enabled product developers to use a knowledge-based ingredient specification baseline

Marketing Data Analyst & Digital Platform Developer, ExxonMobil Jul 2021 – Jul 2021
- Analyzed polymer (VistaMax) physical and chemical properties for performance metrics
- Built and maintained a WeChat Service Platform for supplier communication

- Presented data-driven insights to senior management for strategic decisions

Research & Physician Assistant, Nanjing Drum Tower Hospital Summers 2018, 2019, 2020

- Assisted in cruciate ligament reconstruction surgeries on animal models
- Conducted intra-articular injections, Micro-CT imaging, and DNA sequencing
- Translated orthopedic research papers into English and helped publish in international journals

Research Associate, Jiangsu Kangyuan Pharmaceutical Co. Sep 2019 – Jan 2020

- Operated HPLC to determine purity of "Compound Nanxing Pain Paste" (for osteoarthritis treatment)
- Analyzed physical and chemical properties with mass spectra

TEACHING EXPERIENCE

Graduate Teaching Assistant, Northeastern University Jan 2025 – Present

ME 3475 Fluid Mechanics (Instructors: Prof. Xiaoyu Tang, Prof. Abedi)

- Led recitations and supported delivery of course content, ensuring materials were accurate and accessible to students
- Held regular office hours and review sessions to reinforce core topics (fluid statics, internal/external flows, boundary layers)
- Graded homework, lab reports, quizzes, and exams; provided detailed, actionable feedback to strengthen analytical problem-solving
- Coordinated with instructors to update lecture slides, problem sets, and supplemental resources aligned with course learning objectives
- Facilitated lab sessions by demonstrating experimental setups and guiding data collection and analysis
- Proctored assessments and collaborated with TA team to standardize grading and uphold academic integrity

PROJECT EXPERIENCE

Project Leader (Soft Matter Course), UCSB Mar 2021 – Jun 2021

- Developed a novel non-scrub window cleaner for skyscrapers (1–100 µm dust removal)
- Designed cleaning solution with surfactants, acid components, and hydrophobic coating agents
- Specified ingredient composition, yield strength, and viscosity targets

Team Leader (ChE 110A), UCSB Dec 2019 – Mar 2020

- Designed an automobile gas turbine engine for hybrid electric systems
- Computed standard heat of formation, fuel-air ratio, adiabatic temperature, and thermal efficiency
- Built COMSOL models for Simple/Modified Brayton Cycles, used MATLAB for efficiency/cost mapping

LEADERSHIP & AWARDS

Vice President, Outreach Chair & Treasurer, AIChE (UCSB Chapter) Sep 2018 – Sep 2021

- Hosted environmental restoration projects; supervised volunteering and supply management
- Organized technical workshops & career panels; oversaw budgeting

Member, UCSB Robotics Club Sep 2019 – Dec 2022

- Built elevator-style competition robots for VEX (2020)
- Programmed control modules (C++) for robotic arms and pneumatic gears

Awards:

- **Best Presentation Award**, Research Expo, Northeastern University Nov 2024
- **Champion**, FRC Robotic Challenge, World Adolescent Robot Contest Dec 2016